

EXCLUSIVE

IAS PRELIMS 2026

CURRENT AFFAIRS

REVISION

GS-1 (DYNAMIC)

MODULE - 31



**SCIENCE
&
TECHNOLOGY-7**

How do you define Hunger, Malnutrition, Undernutrition, Wasting and Stunting?

WHAT IS HUNGER?

- It is the proportion of the population, whose food intake is less than its minimum energy requirements. In clearer terms, the food they consume is not sufficient to provide minimum dietary energy requirement for leading a normal, active and healthy life.
- FAO keeps norm of 1,800 kcal. As per ICMR-NIN, the norm is 2,400 kcal for rural areas and 2,100 kcal for urban areas.

WHAT IS MALNUTRITION?

As per UN FAO, 'malnutrition' is the abnormal physical condition caused by unbalanced or excessive intake of macronutrients and/or micronutrients.

WHAT IS UNDERNUTRITION?

Undernutrition is the result of inadequate intake of food, which could be in terms of either quantity or quality, poor utilisation of nutrients due to infections or other illnesses.

WHAT IS STUNTING?

- Those with low height relative to age.
- It is due to chronic undernutrition over a long periods of time. It is irreversible.

WHAT IS WASTING?

- Those with low weight relative to their height.
- It is due to acute undernutrition, over a short-period of time. It is reversible.



- As far as Indian food habits are concerned, diversification is much less, that means we take more rice and wheat and less quantity of pulses, fruits and vegetables.
- Normally many of us do not take millets. Because of lack of diversification, even though we eat sufficient quantity of food, we do not get necessary vitamins and minerals. That's why, we quite often see the deficiency of Iron (anemia), Zinc, Vitamin A, Vitamin D etc.
- Hence, fortifying various foods with these minerals and vitamins is nothing but Food Fortification.

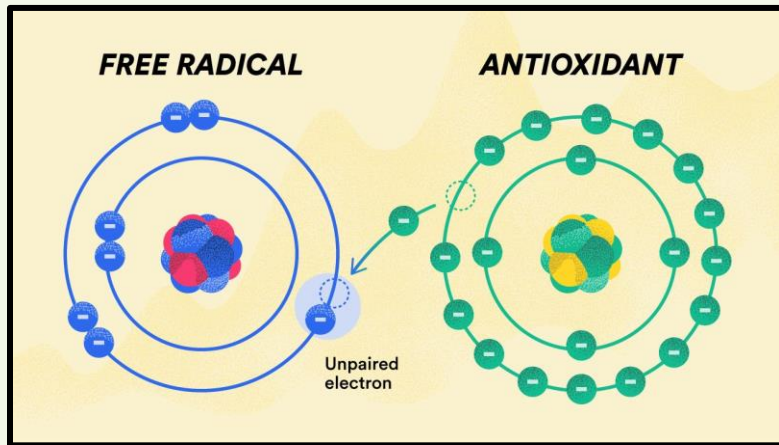
Why do we need Folic Acid, B12 and Iron Fortification?

IAS PRELIMS 2026 S&T



- As per Government, rice supply through Public Distribution System is fortified with three components - Iron, B12 and Folic Acid.
- The significance of these three components :
 - ✓ Iron is stored in our body in the form of Hemoglobin Ferritin. Hemoglobin is oxygen carrying protein, as far as Red Blood Cells are concerned. If Iron is less, we may suffer from iron deficiency anemia, so Iron is important because oxygen carrying depends upon iron.
 - ✓ Folic Acid is the synthetic form of Folate, and it is nothing but B9. Natural folate can be found in foods such as leafy green vegetables, citrus fruits, and beans. Folate plays a crucial role in cell growth and DNA formation. Low folate levels in pregnant women have been linked to birth abnormalities.
 - ✓ Vitamin B12, or cobalamin, is naturally found in animal foods. It can also be added to foods or supplements. Vitamin B12 is needed to form red blood cells and DNA.

What are free radicals? How are they released? What is oxidative stress? What will happen due to continued oxidative stress?



- Free radicals are produced either from normal cell metabolisms i.e. in situ or from external sources. The external sources can be pollution / cigarette smoke / radiation / medication.
- Accumulation of free radicals in the body generates a phenomenon called oxidative stress.
- When oxygen molecules split into single atoms that have unpaired electrons, they become unstable free radicals. If this continues to happen, it begins a process called oxidative stress, which can damage the body's cells, leading to a range of diseases and causes symptoms of aging, such as wrinkles.
- Antioxidants neutralize free radicals. These are either naturally produced in situ or externally supplied through foods and / or supplements.



Why do doctors recommend at least 400-500 grams of fruits and vegetables in our diets daily?

Fruit

Vegetable



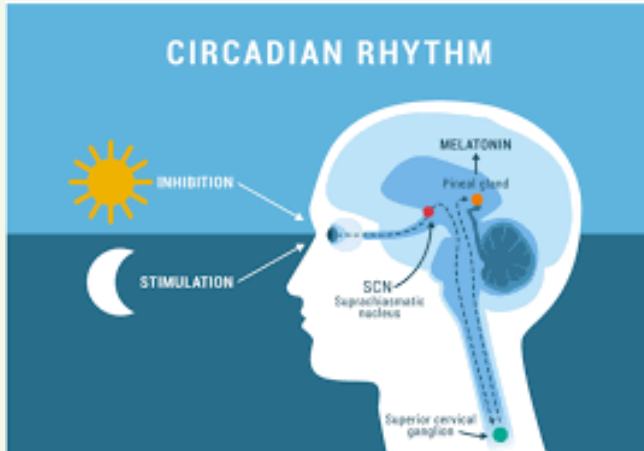
- Fruits and vegetables have lot of fiber, which assists in the digestion process.
- These are antioxidants, which neutralize free radicals. Hence, this prevents damage of the cells by free radicals.
- Fruits and vegetables are also phytochemicals / phytonutrients, which prevents the onset of diseases.

What are Phytochemicals / Phytonutrients?



- These are natural substances produced by plants to protect themselves from fungi, bacteria, viruses, insects, etc. These are found in whole grains, fruits, vegetables, nuts and herbs.
- They can have health promoting effects, eating large amount of food high in phytochemicals may decrease the risk of developing certain cancers as well as diabetes, hypertension and heart disease.
- Foods high in phytochemicals include:
 - ✓ Berries, broccoli, cabbage, cauli flower.
 - ✓ Red, orange and yellow vegetables and fruits (tomatoes, carrots, sweet potatoes, citrus fruits and berries).
 - ✓ Garlic, onions, chives and leeks.
 - ✓ Dark green leafy vegetables.

What do you understand by Circadian Rhythm and Melatonin?



- Circadian rhythm is the 24-hour internal clock in the brain that regulates sleepiness and alertness.
- It is influenced by several factors:
 - ✓ Stress
 - ✓ Physical activity
 - ✓ Social environment
 - ✓ Temperature
 - ✓ Light and dark
 - ✓ Food intake.
- Melatonin is a hormone that our brain produces in response to darkness.
- When we are exposed to light during night time, it blocks melatonin production. Research says that melatonin plays other important roles in the body beyond sleep.



- The concept of UPFs was devised by Carlos Monteiro, a Brazilian scientist, in 2009.
- Mr Monteiro's system, called Nova, puts foods into four buckets:
 - ✓ Unprocessed and minimally processed foods.
 - ✓ Processed culinary ingredients.
 - ✓ Processed foods.
 - ✓ Ultra-Processed foods.
- UPFs, often go through many sophisticated industrial processes. Most contain a blend of artificial ingredients, plenty of salt and sugar and a few nutrients. UPFs are engineered to be easier to eat in large amounts than whole foods.

There is plenty of evidence linking many ingredients in UPFs such as sugar, salt, refined carbohydrates and saturated fats to negative health outcomes.



- Open Digital Ecosystems (ODEs) are open and secure digital platforms that enable a community of actors to unlock transformative solutions for society, based on a robust governance framework.
- It has three layers:
 - ✓ Digital platforms comprising technology, infrastructure and solutions built on top.
 - ✓ Community comprising builders, facilitators and end users.
 - ✓ Layer of governance, which consists of laws and rules and the accountable institutions that uphold them.
- ODEs approach suggests that the Government should focus on creating the digital commons, enable interoperability between siloed systems, so that innovators can build on top, by leveraging open-source software, data, standards and APIs.



The Prime Minister announced National Digital Health Mission with interoperable digital health platform. Efforts like India Stack, DIKSHA and the National Urban Innovation Stack have also adopted a similar approach in other sectors.



- Think tank Indian Software Products Industry Round Table (iSPIRT) unveiled the OCEN framework.
- It is a framework of Application Programming Interfaces for interaction between lenders, loan agents, collection and disbursement partners, derived data providers and account aggregators.
- It is an open network, which codifies the flow of credit between borrowers, lenders and credit distributors under a common set of standards.



What do you understand by Open Network for Digital Commerce (ONDC)?



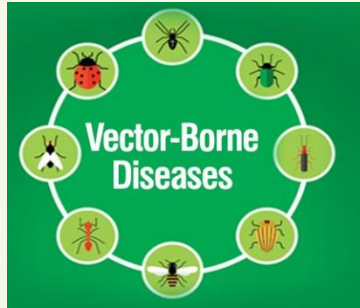
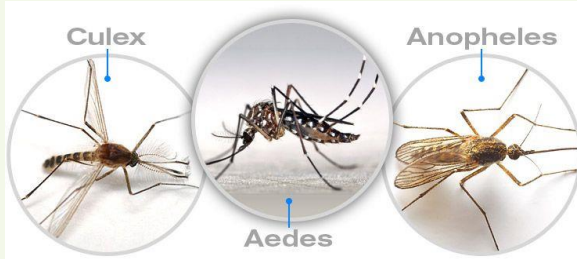
सत्यमेव जयते

Department of Industrial Policy and Promotion
Ministry of Commerce and Industry
Government of India



Open Network for Digital Commerce

- Open Network for Digital Commerce (ONDC) is a public non-profit Section 8 company established by the Department for Promotion of Industry and Internal Trade (DPIIT) to develop open e-commerce.
- It is an initiative aimed at promoting open networks for all aspects of exchange of goods and services over digital or electronic networks.
- It goes beyond the current platform-centric digital commerce model, where the buyer and seller have to use the same platform or application to be digitally visible.
- ONDC protocols would standardize operations like cataloging, inventory management, order management and order fulfillment. It is expected to make e-commerce more inclusive and accessible for consumers. Consumers can potentially discover any seller, product or service by using any compatible application or platform.
- It is expected to promote inclusion of local suppliers, drive efficiencies in logistics and lead to enhancement of value for consumers.



- These are human illnesses caused by parasites, bacteria, virus etc.
- The point to note here is that these are transmitted by vectors, like mosquitoes / ticks.
- The burden is normally high in the tropical and subtropical countries.

SOME OF THESE CARRIERS

- Aedes aegypti mosquito
 - ✓ Chikungunya
 - ✓ Dengue Fever
 - ✓ Yellow Fever
 - ✓ Zika
- Anopheles mosquito
 - ✓ Malaria
 - ✓ Lymphatic Filariasis
- Culex
 - ✓ Japanese Encephalitis
 - ✓ Lymphatic Filariasis



- In the F414 deal, 80% of the cutting-edge technologies are going to be transferred to India. Naturally, this should lead to the diffusion of technologies and knowledge spillovers.
- As far as India is concerned, the ecosystem is not conducive because high-end educational institutions are not many. At the same time, the research and development landscape is not conducive. Hence the knowledge spillovers are difficult in India.
- If we take the case of China, after the establishment of Science Museum in 1980s, the ecosystem of Research Institutions and Educational Institutions flourished.

WE EXPRESS OUR
SINCERE THANKS
FOR VIEWING THIS VIDEO

Presented by



Learning Space

For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

For Information:

info.learningspacedigital@gmail.com

Visit us at:

www.learningspacedigital.com

98499 42299